

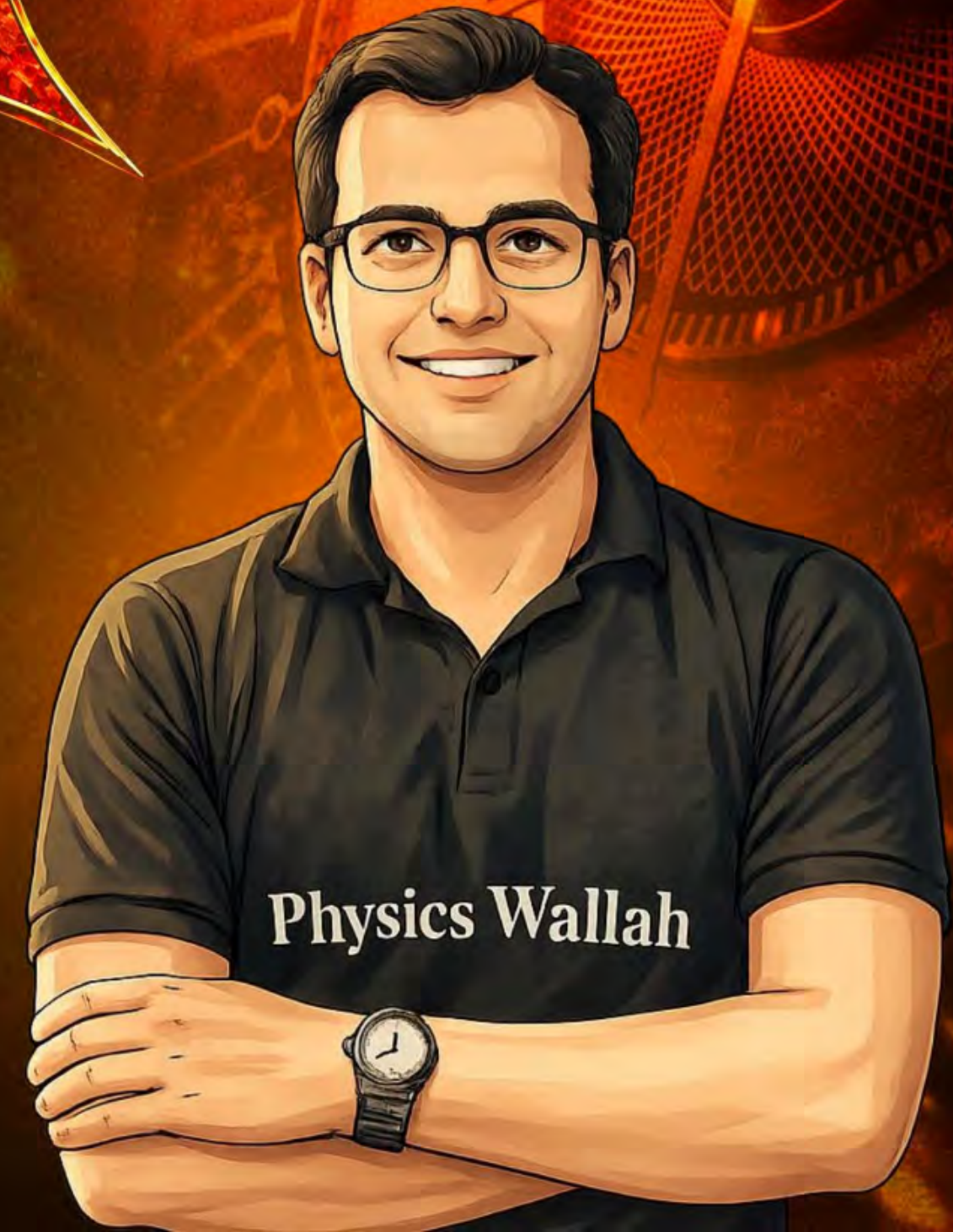
PRAYAS

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Mathematics Lecture-09

Basic Mathematics

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Physics Wallah



Topics to be covered



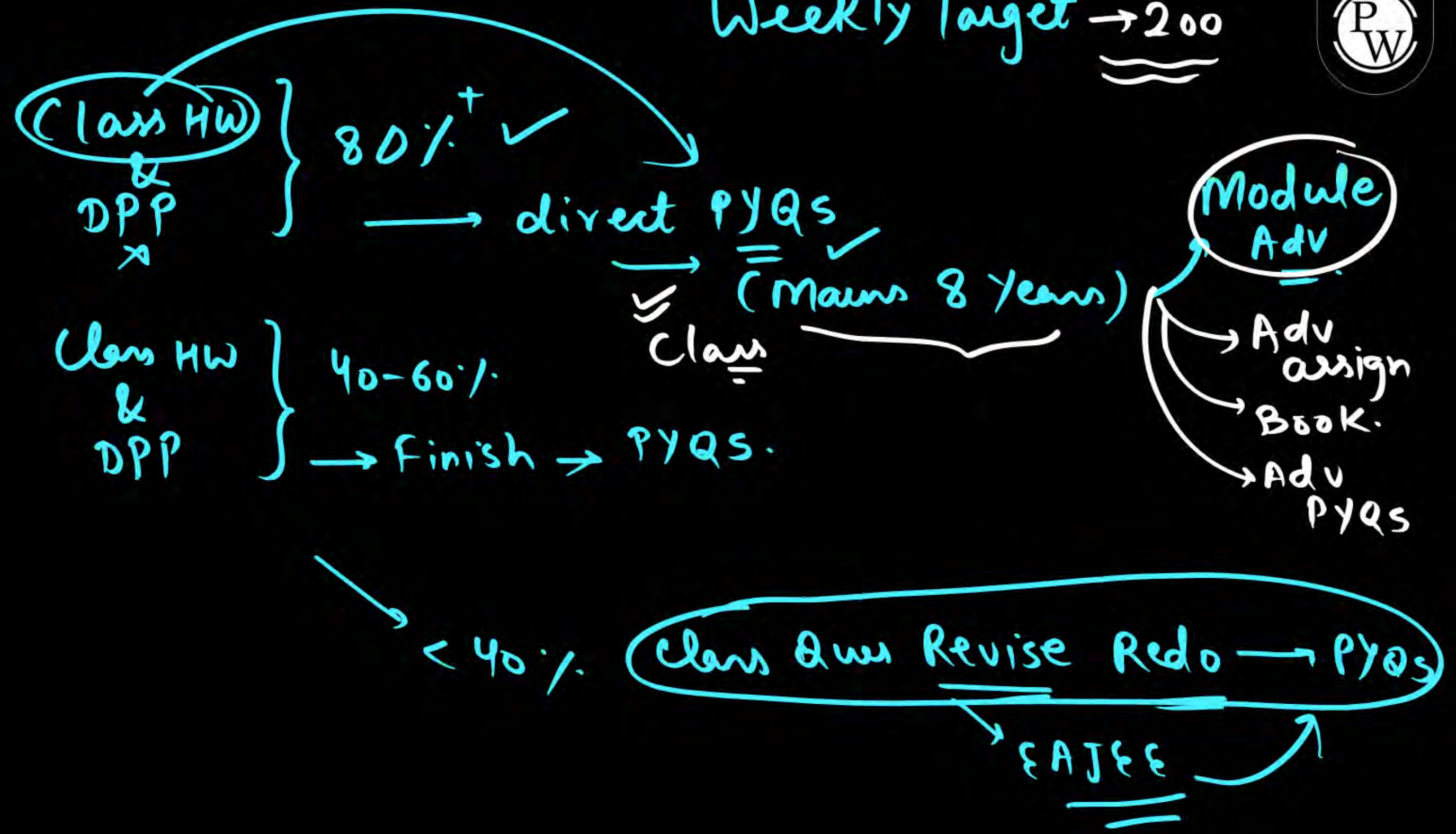
1

Square root inequations – Contd..

2

Logarithm– Properties

Weekly Target $\rightarrow$ 200



$$f(12) = a(12)^3 + b(12)^2 + c(12) + d$$

$$(12)^4 = 20736$$

Case $\rightarrow$ 1 $b=1$

$$12c+d = 352-144$$

$$12c+d = 208$$

$$c+d = 10$$

$$11c+10 = 208$$

$$11c = 198$$

$$c = \frac{198}{11}$$

$$c = 18$$

$$d = -8$$

(X)

Reject

Case $\rightarrow$ 2 $b=2$

$$12c+d = 352-288$$

$$12c+d = 64$$

$$c+d = 9$$

$$11c+9 = 64$$

$$11c = 64-9$$

$$c = \frac{55}{11} = 5$$

$$d = 4$$

$$f(x) = x^3 + 2x^2 + 5x + 4$$

$$f(-1) = 0 \checkmark$$

Find the sum of non real roots of the equation

$$x^5 + 39x^4 + 83x^3 + 325x^2 - 348x - 1924 = 0$$

M-1 $x=2$ & $x=-2$ are roots
 (x^2-4)

M-2

$$x^5 + 83x^3 - 348x + 39x^4 + 325x^2 - 1924 = 0$$

$$x[x^4 + 83x^2 - 348] + 13[3x^4 + 25x^2 - 148] = 0$$

$$x[x^4 + 87x^2 - 4x^2 - 348] + 13[3x^4 + 37x^2 - 12x^2 - 148] = 0$$

$$x[(x^2-4)(x^2+87)] + 13[x^2(3x^2+37) - 4(3x^2+37)] = 0$$

$$x(x^2-4)(x^2+87) + 13(x^2-4)(3x^2+37) = 0$$

$$(x^2 - 4) [x^3 + 87x + 13(3x^2 + 37)] = 0$$

$$x^3 + 87x + 39x^2 + 13 \times 37 = 0$$

$$x^3 + 39x^2 + 87x + 13 \times 37 = 0$$

$$x = -13 \text{ or } x = -37 \checkmark$$

$$(-37)^3 + 39(-37)^2 + 87 \times (-37) + 13 \times 37$$

$$(37)^2 [-37 + 39] + 37 [-87 + 13]$$

$$37 \left[\underbrace{37 \times 2}_{74 - 74} \right] = 0$$

$$x^2(x+37) + 2x(x+37) + 13(x+37)$$

$$(x+37) [x^2 + 2x + 13] = 0$$

$$x^2 + 2x + 13 = 0$$

$$\text{Sum of non real} = -2$$

Brain Teaser

[Ans.: $x \in$

Solve: $\frac{1}{x-1} - \frac{4}{x-2} + \frac{4}{x-3} - \frac{1}{x-4} < \frac{1}{30}$

$$\frac{1}{x-1} - \frac{1}{x-4} + 4\left(\frac{1}{x-3} - \frac{1}{x-2}\right) < \frac{1}{30}$$

$$\frac{x-4 - (x-1)}{(x-1)(x-4)} + 4\left(\frac{x-2 - (x-3)}{(x-3)(x-2)}\right) - \frac{1}{30} < 0$$

$$\frac{-3}{(x-1)(x-4)} + \frac{4 \times 1}{(x-3)(x-2)} - \frac{1}{30} < 0$$

$$t \leftarrow \frac{-3}{\underbrace{x^2 - 5x + 4}} + \frac{4}{x^2 - 5x + 6} - \frac{1}{30} < 0$$

Let $\underline{x^2 - 5x + 4 = t}$

$$-\frac{3}{t} + \frac{4}{t+2} - \frac{1}{30} < 0$$

$$\frac{-3 \times 30(t+2) + 4t(30) - t(t+2)}{t(t+2) \cdot 30} < 0$$

$$\frac{-90(t+2) + 120t - t^2 - 2t}{t(t+2)} < 0$$

$$\frac{-90t - 180 + 120t - t^2 - 2t}{t(t+2)} < 0$$

$$\frac{28t - 180 - t^2}{t(t+2)} < 0$$

$$\frac{t^2 - 28t + 180}{t(t+2)} > 0$$



$$\frac{t^2 - 28t + 180}{t(t+2)} > 0$$

$$\frac{(t-18)(t-10)}{t(t+2)} > 0$$



$$t < -2 \text{ or } 0 < t < 10 \text{ or } t > 18$$

Case 1 $t < -2$

$$x^2 - 5x + 4 < -2$$

$$x^2 - 5x + 6 < 0$$



$$(x-3)(x-2) < 0$$

Case 2 $0 < t < 10$

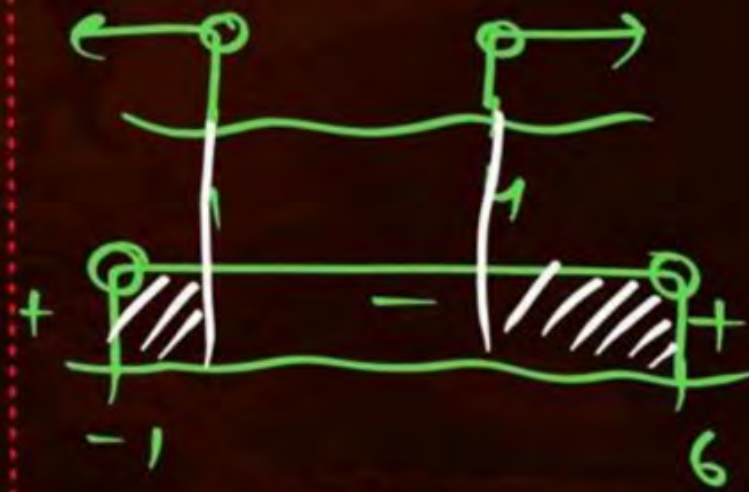
$$0 < x^2 - 5x + 4 < 10$$

$$x^2 - 5x + 4 > 0 \quad \& \quad x^2 - 5x + 4 < 10$$

$$(x-4)(x-1) > 0$$

$$x^2 - 5x - 6 < 0$$

$$(x-6)(x+1) < 0$$



$$x \in (-1, 1) \cup (4, 6)$$

Case 3 $t > 18$

$$x^2 - 5x + 4 > 18$$

$$x^2 - 5x - 14 > 0$$

$$(x-7)(x+2) > 0$$



$$x \in (-\infty, -2) \cup (7, \infty)$$

HW Question

$$\sqrt{8+2x-x^2} > 6-3x$$

[Ans. (1, 4)]



Case → 1
 $RHS > 0 \Rightarrow 6-3x > 0$
 $6 > 3x \Rightarrow \boxed{x \leq 2}$

Case → 2
 $RHS < 0$
 $6-3x < 0 \Rightarrow \boxed{x > 2}$

$$\sqrt{8+2x-x^2} > 6-3x$$

SBS

$$8+2x-x^2 > (6-3x)^2$$

$$8+2x-x^2 > 36+9x^2-36x$$

$$10x^2-38x+28 < 0$$

$$5x^2-19x+14 < 0$$

$$5x^2-5x-14x+14 < 0$$

$$(5x-14)(x-1) < 0$$

LHS = +ve
 RHS = -ve
 LHS > RHS
 always true

$$\boxed{x > 2} \checkmark$$

Step → 1

Domain

$$8+2x-x^2 > 0$$

$$x^2-2x-8 \leq 0$$

$$(x-4)(x+2) \leq 0$$



Case → 1 & Case → 2 ka Union
 $(1, \infty)$

Ans: $(1, 4]$



HW Question

[Ans. (1, 4]]



$$\sqrt{8 + 2x - x^2} > 6 - 3x$$

m-2

By Graph

$$y = \sqrt{8 + 2x - x^2}$$

SBS

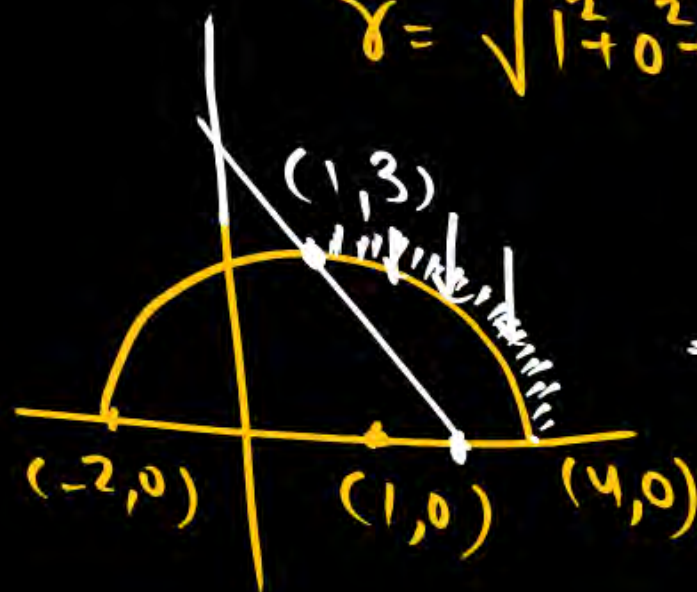
$$y^2 = 8 + 2x - x^2$$

$$x^2 + y^2 - 2x - 8 = 0$$

Semi Circle

(1, 0)

$$r = \sqrt{1^2 + 0^2 + 8} = 3$$



$$\Rightarrow x \in (1, 4]$$

$$y = 6 - 3x$$

SL

x	0	1	2
y	6	3	0

Question

$$\underbrace{\sqrt{x+3} + \sqrt{x+15}}_{+} < \underbrace{6}_{+}$$

SBS Kar Sakte hain

$$(\sqrt{x+3} + \sqrt{x+15})^2 < 36$$

$$x+3 + x+15 + 2\sqrt{x+3}\sqrt{x+15} < 36$$

$$2x + 18 + 2\sqrt{x+3}\sqrt{x+15} < 36$$

$$2\sqrt{(x+3)(x+15)} < 18 - 2x$$

$$\sqrt{(x+3)(x+15)} < 9 - x$$

Case $\rightarrow 1$ RHS < 0

$$9 - x < 0 \Rightarrow \boxed{x > 9}$$

+ve < -ve $\Rightarrow$ Never true $\emptyset$

Case $\rightarrow 2$ RHS ≥ 0

$$9 - x \geq 0$$

$$\boxed{x \leq 9}$$

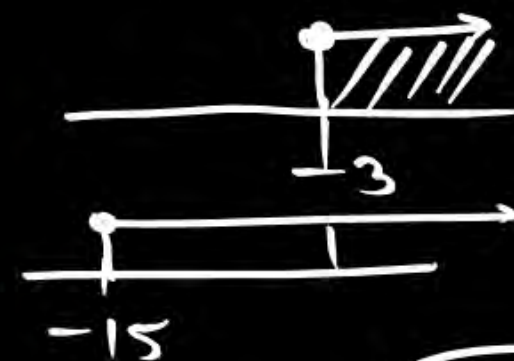
[Ans. $[-3, 1]$]



Domain

$$x+3 \geq 0 \text{ \& } x+15 \geq 0$$

$$x \geq -3 \text{ \& } x \geq -15$$



$$\boxed{x \geq -3}$$

Domain

Case $\rightarrow 2$ $(x \leq 9)$

$$\sqrt{(x+3)(x+15)} < 9-x$$

SBS

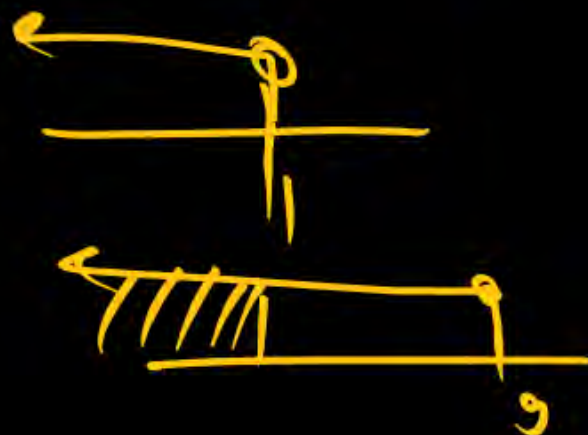
$$(x+3)(x+15) < (9-x)^2$$

$$\cancel{x^2} + 18x + 45 < 81 + \cancel{x^2} - 18x$$

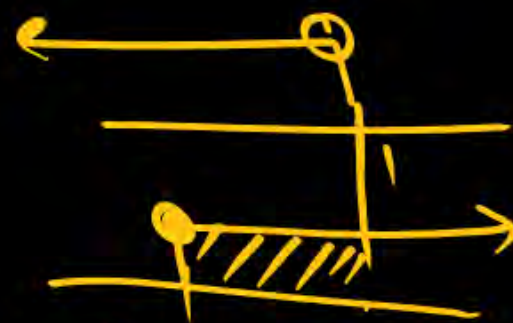
$$36x < 81 - 45$$

$$36x < 36$$

$$\boxed{x < 1}$$



$$\textcircled{1} \cup \textcircled{2} = (-\infty, 1) - 3$$



$$\text{Ans: } x \in (-3, 1)$$

Question

$$\sqrt{x-6} - \sqrt{10-x} \geq 1$$

HW

$$[\text{Ans. } [\frac{16+\sqrt{7}}{2}, 10]]$$





Homework



Solve the following inequalities.

(viii) $\frac{\sqrt{2x^2+15x-17}}{10-x} \geq 0$

(ix) $\sqrt{8+2x-x^2} > 6-3x$

(x) $\sqrt{x+3} + \sqrt{x+15} < 6$

(xi) $\sqrt{2x^2-3x-5} < x-1$

(xii) $\sqrt{x-6} - \sqrt{10-x} \geq 1$

(xiii) $\frac{1-\sqrt{1-4x^2}}{x} < 3$

(xiv) $\sqrt{4-\sqrt{1-x}} - \sqrt{2-x} > 0$



[Ans. (viii) $\left(-\infty, -\frac{17}{2}\right] \cup [1, 10)$

(ix) $(1, 4]$

(x) $[-3, 1)$

(xi) $\left[\frac{5}{2}, 3\right)$

(xii) $\left[\frac{16+\sqrt{7}}{2}, 10\right]$

(xiii) $\left[-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right]$

(xiv) $\left[\frac{\sqrt{13}-5}{2}, 1\right]$



Logarithm-Definition

$$\left(\frac{1}{x}\right)^m = x^{-m}$$



$$\log_b x = y$$

$$x > 0$$

$$b > 0, b \neq 1$$

$\Rightarrow$

$$x = (b)^y$$

learn $\& y \in \mathbb{R}$

$$\log_b 1 = y$$

$$\Rightarrow 1 = b^y$$

$$b^0 = b^y$$

$$\Rightarrow \boxed{y=0} \checkmark$$

$$\log_b 1 = 0$$

Ex $\rightarrow$

① $\log_5 25 = y$
 $5^y = 25 = 5^2$
 $\Rightarrow \boxed{y=2}$

② $\log_5 26 = y$
 $\boxed{26 = 5^y}$
 y is defined

③ $\log_{\frac{1}{\sqrt{2}}} 2 = y$
 $2 = \left(\frac{1}{\sqrt{2}}\right)^y \Rightarrow (\sqrt{2})^2 = (\sqrt{2})^{-y} \Rightarrow \boxed{y=-2}$

Question

Find the value of the following:

$$\log_2 16 = 4$$

$$\log_3 27 = 3$$

$$\log_{1000} 100 = y \Rightarrow 1000^y = 100 \Rightarrow (10^3)^y = (10)^2 \Rightarrow (10)^{3y} = (10)^2 \Rightarrow 3y = 2 \Rightarrow y = 2/3$$

$$2 \log_{10} 100 = 2 \times 2 = 4$$

$$\log_{10} 10^3 = y \Rightarrow 10^y = 10^3 \Rightarrow y = 3$$

$$\log^3_{10} 100 = (\log_{10} 100)^3 = 2^3 = 8$$

$$\log_{10} 100 = 2$$

$$\sin(x^3)$$

$$\sin^3 x = (\sin x)^3$$



Question

$$\text{If } a^2 = b^2 \Rightarrow a = \pm b$$

Find the value of x .

[Ans



$$(1) \log_{(5-x)}(x^2 - 2x + 65) = 2$$

$$x^2 - 2x + 65 = (5-x)^2$$

$$\cancel{x^2} - 2x + 65 = 25 + \cancel{x^2} - 10x$$

$$10x - 2x = 25 - 65$$

$$8x = -40 \Rightarrow \boxed{x = -5}$$

check

$x = -5$ by
Putting in Ques

$\Rightarrow x = -5$ is
accepted

$$(2) \log_{(x-1)}(9) = 2$$

$$\Rightarrow 9 = (x-1)^2$$

$$(x-1)^2 = 3^2$$

$$x-1 = \pm 3$$

$$x = 1 \pm 3$$

$$\checkmark \boxed{x = 4 \text{ or } -2} \rightarrow \text{base} < 0$$

Question

[Ans. (...)]



Find the value of x .

$$(3) \log_3(3^x - 6) = x - 1$$

$$\Rightarrow 3^x - 6 = 3^{x-1}$$

$$3^x - 6 = \frac{3^x}{3}$$

$$\text{Let } 3^x = t$$

$$t - 6 = t/3$$

$$3t - 18 = t$$

$$\Rightarrow 2t = 18$$
$$t = 9 = 3^x$$

$$\boxed{x = 2}$$

$$(4) \log_2(4 + \log_3 x) = 3$$

$$\mu w (x = 81)$$

Question



Solve for x : $\log_4 \left(2 \log_3 (1 + \log_2 (1 + 3 \log_3 x)) \right) = 1/2$

$$\begin{aligned} &\downarrow N \\ \log_4 N &= 1/2 \\ \Rightarrow N &= (4)^{1/2} \end{aligned}$$

$$\boxed{N = 2}$$

$$\log_3 (1 + \log_2 (1 + 3 \log_3 x)) = 1$$

$$1 + \log_2 (1 + 3 \log_3 x) = 3$$

$$\log_2 (1 + 3 \log_3 x) = 2$$

$$3 \log_3 x = 3 \quad 1 + 3 \log_3 x = 4$$

$$x = 3, \log_3 x = 1$$



Basic Properties of Logarithm



$$(P-1) \quad \log_b 1 = 0$$

$$(P-2) \quad \log_a a = 1$$

$$(P-3) \quad \log_{1/a} a = -1$$

$$(P-4) \quad \log_b (xy) = \log_b x + \log_b y$$

$$(P-5) \quad \log_b (x/y) = \log_b x - \log_b y$$

$$\log_b \left(\frac{xyz}{w} \right) = \log_b x + \log_b y + \log_b z - \log_b w$$

$$\begin{aligned} \log_{1/a} a &= y \\ \left(\frac{1}{a} \right)^y &= a = \left(\frac{1}{a} \right)^{-y} \\ y &= -1 \end{aligned}$$

$$\frac{a^x}{a^y} = a^{x-y}$$

$$\begin{aligned} \text{If } \log_b x &= 0 \\ \Rightarrow x &= b^0 = 1 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{Let } \log_a a &= y \\ a &= a^y \\ \Rightarrow y &= 1 \end{aligned}$$

$$\log_\pi \pi = 1$$

$$\log_e e = 1$$

$$\log_1 1 = \text{N.D.} =$$



Basic Properties of Logarithm



$$(P-6) \quad \log_b (x)^y = y \log_b x$$

Proof

$$\begin{aligned} & \log_b (\underbrace{x \cdot x \cdot x \cdot x \cdots}_{y \text{ times}}) \\ &= \underbrace{\log_b x + \log_b x + \log_b x + \cdots}_{y \text{ times}} \\ &= y \log_b x \end{aligned}$$

$$\log_b (1/x) = \log_b (x)^{-1} = -\log_b x$$

$$\log_b 1/x = -\log_b x$$

$$\text{Ex: } \log_2 (2)^{10}$$
$$10 \log_2 2$$

$$10 \times 1 = 10 \checkmark$$

$$\log (1/5) = -\log 5 \checkmark$$



Basic Properties of Logarithm



$$\text{If } \log x + \log y = \log xy$$

$$\textcircled{1} \quad \log_6 2 + \log_6 3 = \log_6 (2 \times 3) = \log_6 6 = 1$$

$$\textcircled{2} \quad \log_6 4 + 2\log_6 3 = ?$$

$$\log_6 4 + \log_6 (3)^2$$

$$\log_6 4 + \log_6 9 = \log_6 (4 \times 9) = \log_6 36 = 2$$



Recap Basic Properties Summary



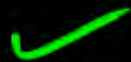
P-1: $\log_a 1 = 0$ i.e., logarithm of unity to any base is zero.



P-2: $\log_N N = 1$ i.e., logarithm of a number to the same base is 1.



P-3: $\log_{\frac{1}{N}} N = -1 = \log_N \frac{1}{N}$ i.e., logarithm of a number to the base as its reciprocal is -1 .



P-4: $\log_b (xy) = \log_b x + \log_b y$

P-5: $\log_b (x/y) = \log_b x - \log_b y$

P-6: $\log_b (x)^y = y \log_b x$

Question



Find the value of:

$$\log_5 [\log_2 \{\log_3 (\log_3 27^3)\}]$$

$$\begin{array}{c} \underbrace{\log_3 (\log_3 27^3)}_9 \\ \downarrow \\ \log_3 9 \\ \underbrace{}_2 \end{array}$$

$$\log_5 (\log_2 2)$$

$$\log_5 (1) = 0 \checkmark$$

$$\log_3 (27)^3$$

$$\begin{aligned} \log_3 (3)^9 &= 9 \log_3 3 \\ &= 9 \end{aligned}$$

A

1

B

2

C

3

D

0 ✓

Question

Simplify the following:



$$243 = 81 \times 3 \\ = 3^4 \times 3 = 3^5$$

$$(i) \log_{10} 2 + \log_{10} 5 = 1$$

$$(ii) \log_{10} 25 + \log_{10} 4 - \log_{10} 10 = \log_{10} \left(\frac{25 \times 4}{10} \right) = \log_{10} 10 = 1 \checkmark$$

$$(iii) \log_{39} \frac{15}{7} + \log_{39} \frac{13}{3} - \log_{39} \frac{5}{21} = \log_{39} \left(\frac{15}{7} \times \frac{13}{3} \div \frac{5}{21} \right) = \log_{39} \left(\frac{15}{7} \times \frac{13}{3} \times \frac{21}{5} \right) = 1$$

$$(iv) \log_3 243 = \log_3 (3)^5 = 5 \times 1 = 5$$

$$(v) 2 \log_6 2 + 3 \log_6 3 + \log_6 12 = \log_6 (2^2 \times 3^3 \times 12) = \log_6 (2^2 \cdot 3^3 \cdot 2^2 \cdot 3) = \log_6 (2^4 \cdot 3^4) = 4 \log_6 6 = 4$$

Examples on Basic Properties of Logarithm (P-1 to P-6):

[Ans. (A)]



Question

The value of $7 \log \frac{16}{15} + 5 \log \frac{25}{24} + 3 \log \frac{81}{80} = ?$

HW

- A** $\log 2$
- B** $\log 3$
- C** $\log 5$
- D** None of these



Mistakes While Applying Log



Wrong

$$1) \frac{\log_b x}{\log_b y} = \log_b x - \log_b y$$

$$2) \log x \cdot \log y = \log x + \log y$$

$$3) \log(x+y) \begin{cases} \rightarrow \log x + \log y \\ \rightarrow \log(xy) \end{cases}$$

Correct

$$\log\left(\frac{x}{y}\right) = \log x - \log y$$

$$\log(xy) = \log x + \log y$$

$$\log(x+y) \rightarrow \text{no property}$$

$$\log(x-y) \rightarrow$$

Examples on Basic Properties of Logarithm (P-1 to P-6):

[Ans. (-1)]



Question

Simplify: $\log_{1/3} \sqrt[4]{729 \cdot \sqrt[3]{9^{-1} \cdot 27^{-4/3}}}$

HW

Examples on Basic Properties of Logarithm (P-1 to P-6):

Question

HW

	Column-I		Column-II
(1)	$\frac{\log_3 243}{\log_2 \sqrt{32}}$	(A)	positive integer
(2)	$\frac{2\log 6}{(\log 12 + \log 3)}$	(B)	negative integer
(3)	$\log_{1/3} \left(\frac{1}{9}\right)^{-2}$	(C)	rational but not integer
(4)	$\frac{\log_5 16 - \log_5 4}{\log_5 128}$	(D)	prime

Question

The value of $\sqrt{(\log_{0.25}^2(16))}$ is:

- A** -2
- B** $\sqrt{-4}$
- C** 2 ✓
- D** None of these

$$0.25 = \frac{1}{4}$$

$$\log^2(x) = (\log x)^2$$

[Ans. (C)]



Mistake

$$\log_{1/4} 16$$

$$\log_{1/4} (4)^2$$

$$2 \log_{1/4} 4$$

$$2 \times (-1)$$

$$= -2$$

Correct

$$|\log_{1/4} 16|$$

$$|-2| = 2$$

$$\sqrt{x^2} = |x|$$

Example

H/W

[Ans. (C)]



Solve for x : $\log_{135} \{(2^x - 7) + \log_{135}(2^x - 1)\} = 1$

- A** -8 or 16
- B** 0 or 4
- C** 4 Only
- D** None of these



Two Most Common Bases of Logarithm



$$\log_b x$$

$b > 0, b \neq 1$

$b=10$
Common log
 $\log_{10} x$

base = e [Napier's const] $e \approx 2.71$
or Euler's const

$$\log_e x = \ln x$$

Natural log

$$\ln(RY) = \ln R + \ln Y$$

$\downarrow \quad \quad \downarrow$
 $P \quad \quad W$

$$\ln P + \ln W$$

$\downarrow$
 $\ln(PW)$

$$\ln e = 1$$

$$\ln e^2 = 2$$

$$\ln \frac{1}{e} = -1$$

$$\ln 1 = 0$$

$$\text{If } \ln(x) = \ln(y) \Rightarrow x = y$$

$$\text{If } \ln x = \ln y + \ln z \Rightarrow \ln x = \ln yz \Rightarrow x = yz$$

$$\text{If } \ln x = 2 \ln y \Rightarrow x = y^2$$

$$\text{If } \ln x = \ln y + 2 \ln z$$

$$\Rightarrow \ln x = \ln(y z^2)$$

$$\Rightarrow \boxed{x = y z^2}$$

Question

[Ans. (A)]



If $\log_e \left(\frac{a+b}{2} \right) = \frac{1}{2} (\log_e a + \log_e b)$, then relation between a and b will be:

- A** $a = b$
- B** $a = b/2$
- C** $a = 2b$
- D** $a = 2b$

$$\ln \left(\frac{a+b}{2} \right) = \frac{1}{2} \ln(ab)$$

$$\ln \left(\frac{a+b}{2} \right) = \ln (ab)^{1/2}$$

$$\frac{a+b}{2} = (ab)^{1/2}$$

SBS

$$\frac{(a+b)^2}{4} = ab$$

$$a^2 + b^2 + 2ab = 4ab$$

$$a^2 + b^2 - 2ab = 0$$
$$a = b \quad (a-b)^2 = 0$$

Seq & Series

$$AM = GM$$
$$\Rightarrow \{a=b\}$$

m-2

Brain Teaser

[Ans. (B)]



The number of solutions of the equation $\log(-x) = 2 \log(x+1)$, is:

A 0 ✗

B 1 ✓

C 2 ✗

D 4

$$\log(-x) = \log(x+1)^2$$

$$\Rightarrow -x = (x+1)^2$$

$$-x = x^2 + 1 + 2x$$

$$x^2 + 3x + 1 = 0$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4}}{2 \times 1} = \frac{-3 \pm \sqrt{5}}{2}$$

$$x = \frac{-3 - \sqrt{5}}{2}$$

$$\text{or } x = \frac{-3 + \sqrt{5}}{2}$$

$$x+1 = \frac{-3 + \sqrt{5}}{2} + 1 = \frac{-1 + \sqrt{5}}{2} \rightarrow \text{+ve}$$

Reject

$$x+1 = \frac{-3 - \sqrt{5}}{2} + 1 = \frac{-1 - \sqrt{5}}{2} \rightarrow \text{-ve}$$

Accept

Question

Solve for x : $x^{(\log_{10} x)+1} = 10^6$

[Ans. ($x = 100$ OR $1/1000$)]



taking $\log_{10}$ both sides

$$\log_{10} (x)^{\log_{10} x + 1} = \log_{10} (10)^6$$

$$\log x + 1$$

$$\log(x+1)$$

$$(\log_{10} x + 1) \log_{10} x = 6 \times 1$$

$$\text{Let } \log_{10} x = t$$

$$(t+1)t = 6$$

$$t^2 + t = 6$$

$$t^2 + t - 6 = 0$$

$$t = 2 \text{ or } -3$$

$$\begin{aligned} t^2 + 3t - 2t - 6 &= 0 \\ t(t+3) - 2(t+3) &= 0 \\ (t+3)(t-2) &= 0 \end{aligned}$$

$$\log_{10} x = 2 \text{ or } \log_{10} x = -3$$

$$\Rightarrow x = 10^2 \text{ or } x = 10^{-3}$$

$$x = 100 \text{ or } x = \frac{1}{1000}$$

Question

Solve for x : $x^{2 \log x} = 10x^2$

(where base of log is 10)

HW $\log_{10}$ both sides.



Question

[Ans. (A, B, C)]



Indicate all correct alternatives, where base of the log is 2.

The equation $x^{\left[\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}\right]} = \sqrt{2}$ has:

- A** Exactly 2 Rational solution ✓✓
- B** Exactly 3 real solutions ✓✓
- C** Exactly one irrational solution ✓✓
- D** Non-Real Solutions

$\log_2$ both sides

$$\left(\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}\right) \log_2 x = \log_2 \sqrt{2}$$

Let $\log_2 x = t$

$$\left(\frac{3}{4}t^2 + t - \frac{5}{4}\right)t = \frac{1}{2}$$

$$\frac{(3t^2 + 4t - 5)t}{4} = \frac{1}{2}$$

$$3t^3 + 4t^2 - 5t = 2$$

↓
 $\log_2 (2)^{1/2}$

$$\frac{1}{2} \times 1 = \frac{1}{2}$$

$$\boxed{3t^3 + 4t^2 - 5t - 2 = 0} \checkmark$$

$$t = 1 \text{ (HnT)}$$

$$3t^2(t-1) + 7t(t-1) + 2(t-1)$$

$$(3t^2 + 7t + 2)(t-1) = 0$$

$$(3t^2 + 6t + t + 2)(t-1) = 0$$

$$(3t+1)(t+2)(t-1) = 0$$

$$t = 1 \text{ or } -1/3 \text{ or } -2$$

$$\log_2 x = 1 \text{ or } -1/3 \text{ or } -2$$

$$x = 2^1 \text{ or } 2^{-1/3} \text{ or } 2^{-2}$$

$\downarrow$ $\downarrow$ $\downarrow$
 2 $2^{1/3}$ $2^{1/4}$

Question

HW

[Ans. (A, B, C)]



The equation $x^{[(\log_3 x)^2 - \frac{9}{2} \log_3 x + 5]} = 3\sqrt{3}$ has:

- A** Exactly 3 real solutions
- B** 2 Integral Solutions
- C** Exactly one irrational solution
- D** Non-Real Roots

Brain Teaser

HW



Solve the system of equations for x , y and z :

$$(\log_{2025} x) \log_{2025}(xyz) = 48$$

$$(\log_{2025} y) \log_{2025}(xyz) = 12$$

$$(\log_{2025} z) \log_{2025}(xyz) = 84$$



Advanced Properties of Logarithm



P-7:

Base Change Theorem

$$\log_b N = \frac{\log_c N}{\log_c b}$$

log of Number
log of base

$$\text{Ex} \rightarrow \log_3 7 = \frac{\log_2 7}{\log_2 3} = \frac{\log_e 7}{\log_e 3}$$

$$\log_e 10 = \frac{\log_2 10}{\log_2 e}$$

Question



The value of $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$ is:

A 1

B 2 ✓

C 3

D 4

$$\frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times \frac{\log 6}{\log 5} \times \frac{\log 7}{\log 6} \times \frac{\log 8}{\log 7} \times \frac{\log 9}{\log 8}$$

$$\frac{\log 9}{\log 3} = \frac{2 \log 3}{\log 3}$$

$$= 2$$

Question

HW

[Ans. (511)]



Find the value of $\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \dots \log_n (n+1) = 9$ find n .



Advanced Properties of Logarithm



Derived properties of Base Change Theorem

P-8:

$$\log_b a = \frac{1}{\log_a b}$$

$$a, b \neq 1$$

or

$$(\log_b x)(\log_x b) = 1$$

$$\text{Ex: } \log_3 10 = \frac{1}{\log_{10} 3}$$

$$\log_e \pi = \frac{1}{\log_\pi e}$$

$$\log_{20} 10 = \frac{1}{\log_{10} 20}$$

Question



Solve the equation $1 + 2\log_{(x+2)} 5 = \log_5 (x + 2)$

$\downarrow$ t $\downarrow$ $\frac{1}{t}$

$$1 + 2t = \frac{1}{t} \quad \checkmark$$

Brain Teaser

Solve the equation $\log_x (9x^2) \cdot (\log_3 x)^2 = 4$

HW



Question

Find the Exact value of $\frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$

HW

[Ans. (3)]





Homework



❖ **DPP- 07**

Thank

You